AOI
ARCHITECTURE OF INTEGRITY

FW-001 FRAMEWORK SERIES

THE RUNTIME GOVERNANCE DOMAIN
IN THE AI RISK ECOSYSTEM

Existing disciplines govern identity, authorization, security, AI risk, and continuous trust. Runtime Governance evaluates whether autonomous execution remains a legitimate exercise of enterprise authority.

REV 02.0
AUGUST 2026
AOI FRAMEWORK SERIES

FRAMEWORK PURPOSE:

This framework defines the Runtime Governance domain within the enterprise AI risk ecosystem. It clarifies how autonomous execution changes the object of governance and why a distinct discipline—Runtime Governance—is required to evaluate whether execution remains a legitimate exercise of enterprise authority.

Rather than asking only, "Is this principal permitted to perform this operation?" Runtime Governance asks, "Is this specific execution still a legitimate exercise of the authority that was granted?"

DATA RISKS

- **BS** Data quality and accuracy risk
- **PV** Privacy and sensitivity risk
- **LE** Data leakage exposure
- **SD** Synthetic data risk
- **SE** Embedded secrets risk

Focus: Data as an asset

MODEL RISKS

- **HL** Hallucination and output risk
- **EX** Explainability and transparency risk
- **OF** Overfitting and bias risk
- **MD** Model drift and degradation risk
- **PS** Prompt sensitivity and injection risk

Focus: Model behavior

AGENT RISKS

- **AH** Autonomy and goal alignment risk
- **TM** Tool misuse and abuse risk
- **UA** Unintended actions risk
- **PE** Permission escalation risk
- **IN** Looping and peristence risk
- **SM** Social manipulation and influence risk
- **DG** Delegation misuse risk
- **MR** Memory and retention risk
- **LF** Long failure repeats risk
- **RG** Rollback and graph complexity risk

Focus: Agent behavior

SECURITY & OPS RISKS

- **VP** Vulnerabilities and exposures
- **CT** Cyber threats and attack surface
- **AB** API abuse and misuse
- **TT** Token theft and abuse
- **MU** Malicious use and intent
- **DP** Deployment risk and unsafe config
- **IG** Integration risk and third-party
- **LT** Latency and availability risk
- **CO** Cost overrun and resource risk

Focus: Infrastructure and operations

GOVERNANCE RISKS

- **MG** Monitoring gap and blind spots
- **DO** Decision and ownership opacity
- **HO** Human oversight gap
- **TR** Traceability and attribution gap
- **CP** Compliance risk and standards misalignment
- **AG** Audit gap and evidence insufficiency
- **LG** Legal risk and IP and copyright
- **IP** risk
- **ET** Ethics and harm risk
- **RR** Regulatory risk and rule changes

Focus: Governance of systems

RUNTIME GOVERNANCE
(THE RUNTIME GOVERNANCE DOMAIN)

Evaluates whether autonomous execution remains a legitimate exercise of enterprise authority.

1. **AUTHORITY VALIDITY** — Is every authority in the chain currently valid?
2. **AUTHORITY PROVENANCE** — Does the action trace to authority actually conferred—and does the chain remain within that authority?
3. **CONTEXTUAL VALIDITY** — Do the conditions that justified the authority still exist?
4. **INTENT CONFORMANCE** — Does the action remain within the objective that initiated the execution?
5. **CAUSAL INTEGRITY** — Did the action originate from legitimate principal intent (not from data, content, or other indirect instruction)?

Focus: Execution legitimacy

THE EXECUTIVE REALITY

An execution can be:

- ☑ Authenticated
- ☑ Currently authorized
- ☑ Policy compliant
- ☑ Fully logged
- ☑ Technically correct

AND STILL FAIL EXECUTION LEGITIMACY.

THE QUESTION THAT MATTERS MOST

WAS THIS EXECUTION A LEGITIMATE EXERCISE OF THE AUTHORITY THAT WAS GRANTED?

SECURITY CORRECTNESS — Systems behaved according to configured rules.

≠

EXECUTION LEGITIMACY — The execution satisfied the five legitimacy conditions when it occurred.

A SYSTEM MAY BE ENTIRELY CORRECT AND ENTIRELY ILLEGITIMATE AT THE SAME TIME.

WHY IT MATTERS

- Prevents unauthorized actions with valid credentials.
- Reduces risk from stale, diverged, or delayed state.
- Strengthens audit, traceability, and compliance.
- Protects people, data, IP, and the organization.
- Creates independently verifiable governance evidence.

THE EVOLUTION OF THE GOVERNANCE OBJECT

NETWORKS	PRINCIPALS	REQUESTS	AUTONOMOUS EXECUTION
Governed communication between systems.	Governed who could access resources.	Governed whether requests were permitted.	Governed whether execution remains a legitimate exercise of conferred authority.
DISCIPLINE Network Security →	DISCIPLINE Identity Governance →	DISCIPLINE Zero Trust / Continuous Authorization →	DISCIPLINE Runtime Governance

ABOUT RUNTIME GOVERNANCE

Runtime Governance evaluates whether a specific execution event constitutes a legitimate exercise of conferred authority at the moment it occurs, and produces independently verifiable evidence of what those evaluations concluded.

It is not a replacement for existing disciplines. It is an extension that addresses a different evaluation object: the execution event.

EXECUTION EVENT
↓
LEGITIMACY EVALUATION
↓
GOVERNANCE EVIDENCE

RELATED RUNTIME GOVERNANCE CONCEPTS

- **EXECUTION EVENT** — The composite object evaluated by Runtime Governance.
- **DELEGATION CHAIN** — The ordered sequence through which authority is conveyed.
- **AUTHORITY DRIFT** — The divergence between authority held and what current circumstances justify.
- **GOVERNANCE EVIDENCE** — The reasoning and inputs supporting a legitimacy determination.
- **DECISION RECEIPT** — The atomic evidence record of a governance evaluation.

GOVERNANCE LAW

SECURITY CAN BE CORRECT. | AUTHORITY CAN BE VALID. | EXECUTION CAN STILL BE ILLEGITIMATE. | Runtime Governance evaluates the difference.